

## **WEEK 01: TEAM INFORMATION SHEET**

|                    |                                                                                                                                 |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Official Team Name | CSE4204-8A-T03                                                                                                                  |
| Section            | 8A                                                                                                                              |
| Project Title      | AI-Based Smart Quiz System                                                                                                      |
| GitHub Repository  | <a href="https://github.com/sudipto224/AI-based-Smart-Quiz-System">https://github.com/sudipto224/AI-based-Smart-Quiz-System</a> |

### **Team Information Sheet:**

| <b>Role</b> | <b>Full Name</b>      | <b>Student ID</b> |
|-------------|-----------------------|-------------------|
| Team Leader | Sudipto Das           | 11220320830       |
| Member 02   | Chaon Das             | 11210320629       |
| Member 03   | Rakibur Rahman Mishan | 11220320829       |
| Member 04   | Aditi Roy             | 11220320846       |

### **Preliminary Project Description:**

The AI-Based Smart Quiz System is a web-based online assessment platform built with Laravel. It provides three quiz difficulty levels (Easy – 8 questions, Medium – 10 questions, Hard – 10 questions) with a per-question countdown timer (default 30 seconds, auto-submit on timeout). After submission, students receive their score, a list of wrong answers, correct answers, and pre-stored explanations. A leaderboard ranks all users by highest score and shortest completion time. Administrators have a dedicated panel to add, view, and delete questions (each question stores level, text, four options, correct answer, and explanation).

### **Two AI-powered features are being integrated:**

**AI Question Generation:** Admin enters a topic; AI (Gemini/OpenAI) automatically creates a full MCQ (question, 4 options, answer, explanation) and saves it to the database.

**Cheating Detection:** Real-time monitoring of tab switches (`visibilitychange` event) and flagging of suspiciously fast submissions (average <5 seconds per question). Suspicious attempts are marked for admin review.

### AI is Needed in the Project:

Manual question creation is slow and repetitive, while online quizzes often lack integrity due to tab-switching or extremely fast submissions. Existing generic platforms (Google Forms, Kahoot, etc.) offer no intelligent content generation nor real-time cheating detection.

#### AI solves these problems by:

- Automating question creation – reducing hours of manual work to seconds.
- Enhancing exam integrity – detecting tab switches and abnormal submission speeds, flagging potential cheating for administrative review.
- Scalability – enabling dynamic, fresh question banks without extra human effort.
- Thus, AI makes the system efficient, fair, and scalable.

### Initial Technology Stack:

| Category        | Technologies                          |
|-----------------|---------------------------------------|
| Frontend        | HTML5, Custom CSS, Vanilla JavaScript |
| Backend         | Laravel 12 (PHP)                      |
| Database        | MySQL / PostgreSQL                    |
| AI Tools        | Google Gemini API / OpenAI API        |
| Authentication  | Laravel Breeze                        |
| Version Control | Git & GitHub                          |

### Similar Existing Systems / Research:

- **Kahoot!** – Gamified quizzes, but no AI question generation or cheating detection.
- **Google Forms** – Basic quiz mode, no timer, no AI, no leaderboard.
- **Moodle** – Powerful but complex; requires manual question entry; limited AI integration.
- **Quizlet** – Flashcard style, not timed assessment; no cheating detection.

- **Research:** Several papers propose using GPT models for automatic MCQ generation (e.g., “Automatic Question Generation using Transformers”); our system extends this by integrating real-time proctoring features.